EDS Practical :04

Name: Shrusti Dnyandev Sontakke

PRN: 202501120039

Department: Data science

Division: DS 1

Batch: DS 12

4.1.1 Pandas - series creation and manipulation:

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and

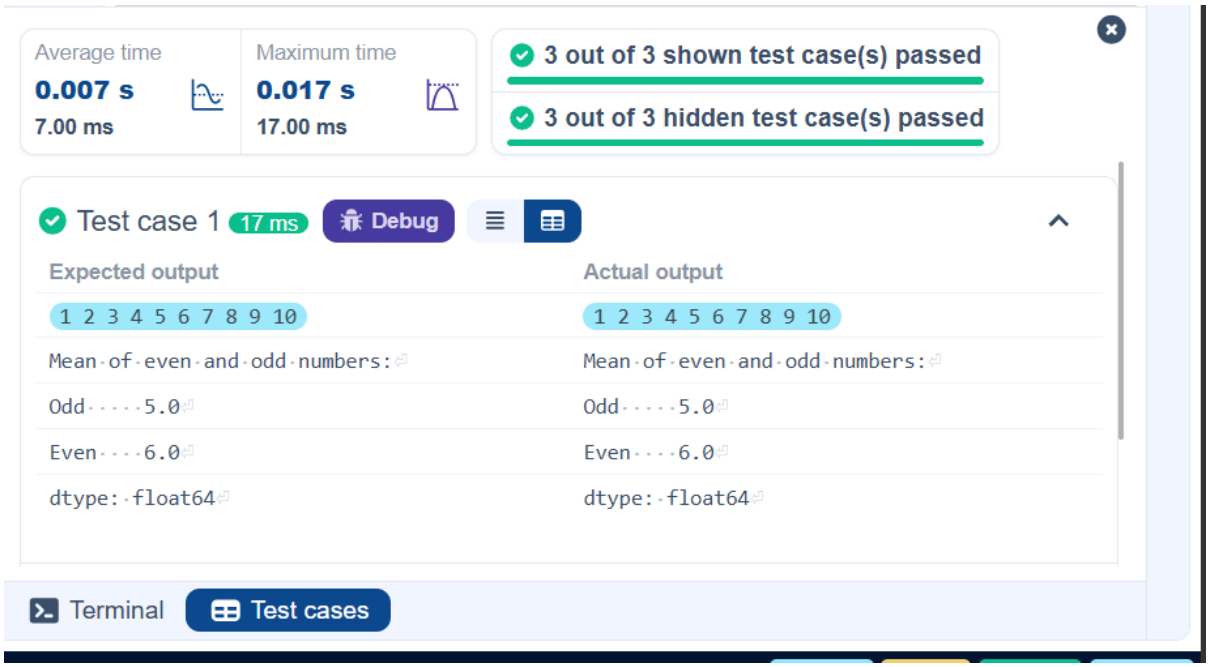
odd numbers separately.

- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Code:

```
import pandas as pd

# Take inputs from the user to create a list of numbers
numbers = list(map(int, input().split()))

# Create a Pandas series from the list of numbers
series = pd.Series(numbers)
# Grouping by even and odd numbers and calculating the mean
grouped =series.groupby(series % 2 == 0).mean()

# Display the mean of even and odd numbers with labels
grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
print("Mean of even and odd numbers:")
print(grouped)
```

Output:



4.1.2. Dictionary to dataframe:

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.

- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column Gender with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the Age column.
- Display the DataFrame after deleting the column.

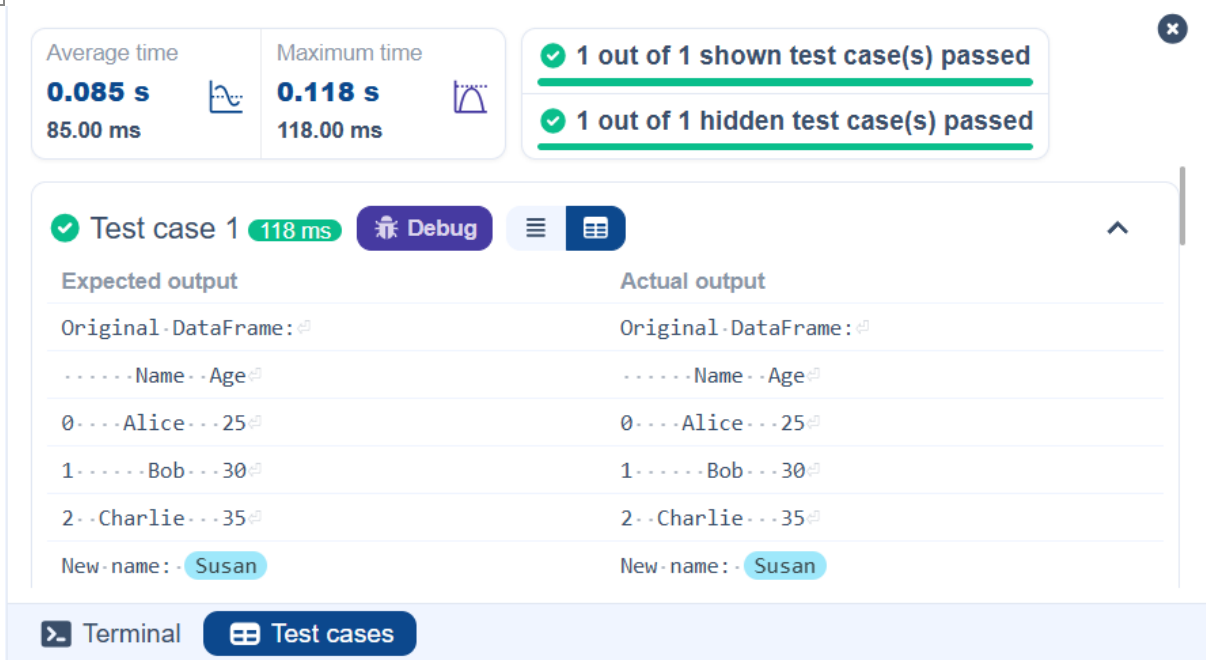
Code:

```
import pandas as pd

# Provided dictionary of lists
data = {
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35],
}

# Convert the dictionary to a DataFrame
df = pd.DataFrame(data)

# Display the original DataFrame
print("Original DataFrame:")
print(df)

# Adding a new row
new_name = input("New name: ")
```

```
new_age = int(input("New age: "))
df.loc[len(df)] = [new_name, new_age]
# Display the DataFrame after adding a new row
print("After adding a row:\n",df)

# Modifying a row
row_to_modify = int(input("Index of row to modify: "))
new_age_value = int(input("New age: "))
df.at[row_to_modify, "Age"] = new_age_value
# Display the DataFrame after modifying a row
print("After modifying a row:")
print(df)

# Deleting a row
row_to_delete = int(input("Index of row to delete: "))
df =
df.drop(index=row_to_delete).reset_index(drop=True)
# Display the DataFrame after deleting a row
print("After deleting a row:")
print(df)
```

```
# Adding a new column
genders=input("Enter genders separated by space: ").split()
df["Gender"] = genders

# Display the DataFrame after adding a new column
print("After adding a new column:")
print(df)

# Modifying a column
df["Name"]=df["Name"].str.upper()
# Display the DataFrame after modifying a column
print("After modifying a column:")
print(df)

# Deleting a column
df=df.drop(columns=["Age"])
# Display the DataFrame after deleting a column
print("After deleting a column:")
print(df)
```

Output:



4.1.3. Student Information:

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

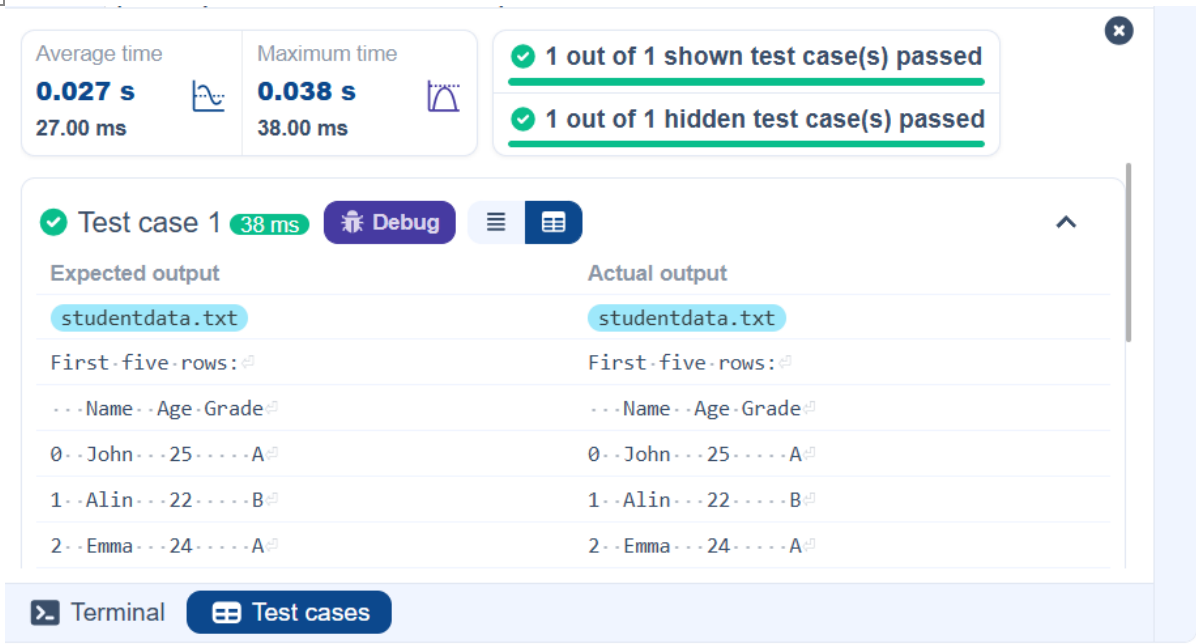
- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Code:

```
import pandas as pd
```

```
# Read the text file into a DataFrame
file = input()
data = pd.read_csv(file, sep="\s+", header=None,
names=["Name", "Age", "Grade"])



# write your code here..
df=pd.DataFrame(data)
print("First five rows:")
print(df.head(5))
print("Average age:",df["Age"].mean().round(2))
print("Students with a grade up to B")
filtered_df=df[df['Grade'].isin(['A','B'])]
print(filtered_df)
```

Output:



4.2.1. Month with the Highest Total Sales:

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

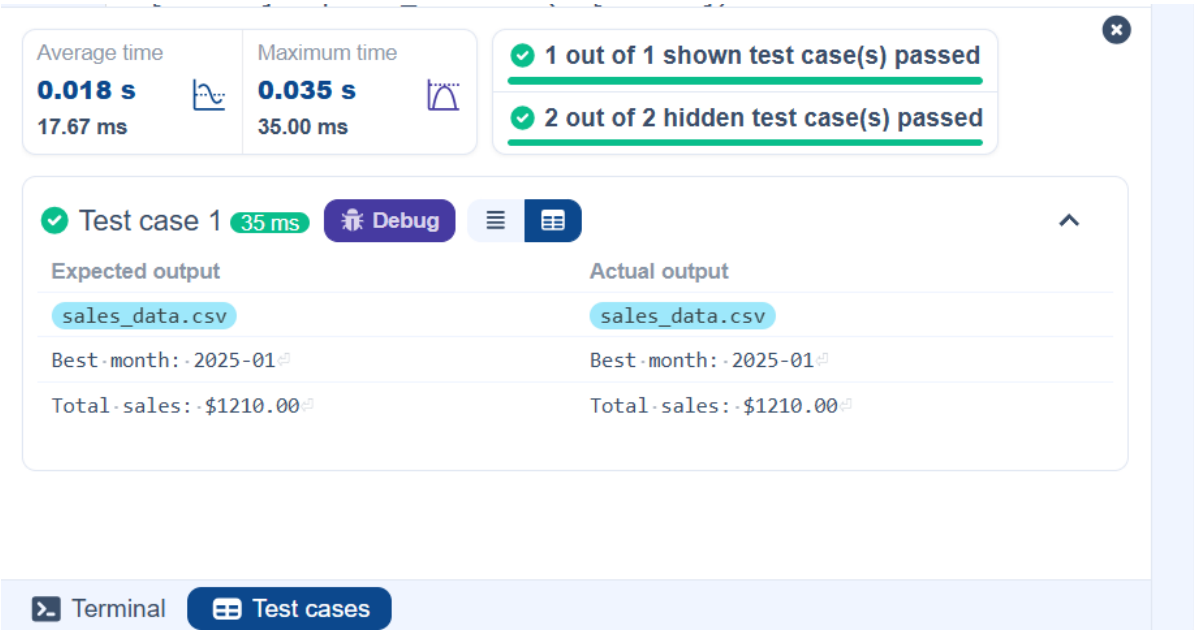
Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code:

```
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)
df["Date"] = pd.to_datetime(df["Date"])

df["Month"] = df["Date"].dt.to_period("M")

df["Total Sales"] = df["Quantity"] * df["Price"]

monthly_sales = df.groupby("Month")["Total
Sales"].sum()
# Find the month with the highest total sales
best_month = monthly_sales.idxmax()
highest_sales = monthly_sales.max()

print(f"Best month: {best_month}")
print(f"Total sales: ${highest_sales:.2f}")
```

Output:



4.2.2. Best Selling Product:

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

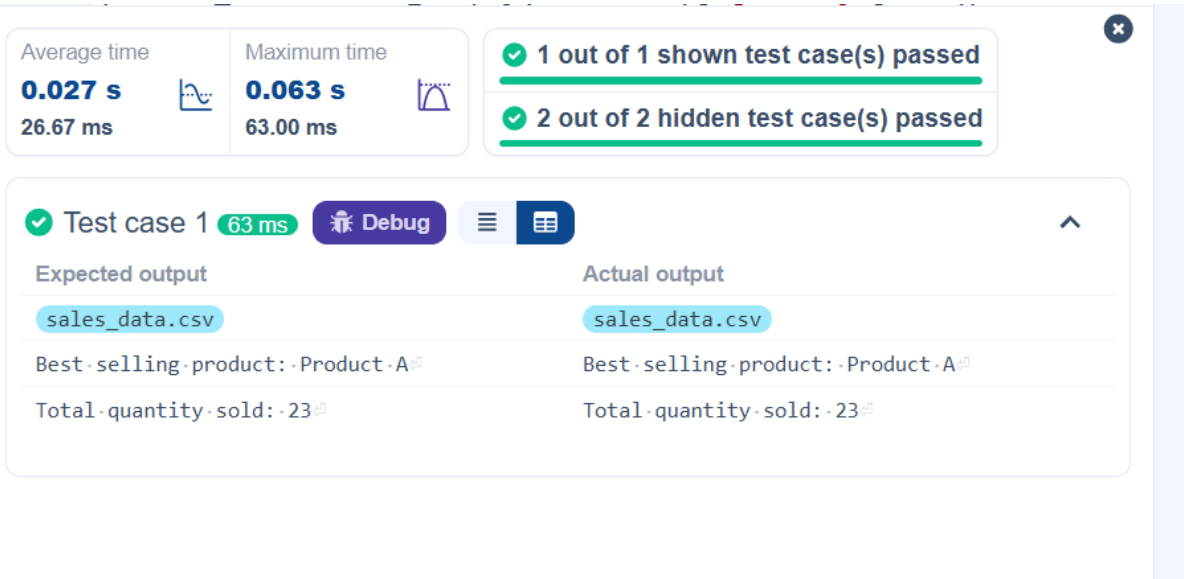
- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Code:

```
import pandas as pd
```

```
# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

product_sales = df.groupby("Product")["Quantity"].sum()
# Find the product with the highest total quantity sold
best_product = product_sales.idxmax()
highest_quantity = product_sales.max()

# Display the result
print(f"Best selling product: {best_product}")
print(f"Total quantity sold: {highest_quantity}")
```

Output:



4.2.3. City that Sold the Most Products:

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

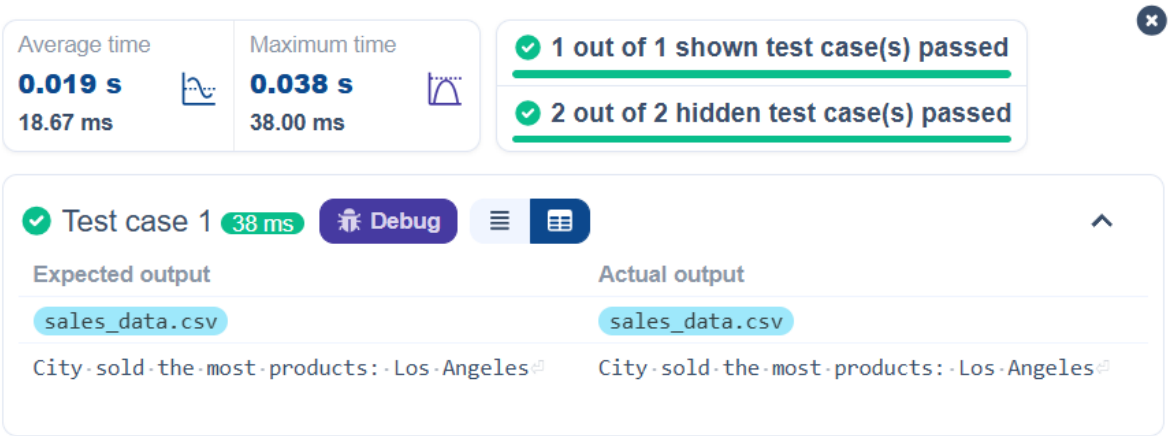
- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Code:

```
import pandas as pd


# Prompt the user for the file name
```

```
file_name = input()

# Load the data
df = pd.read_csv(file_name)

city_sales = df.groupby("City")["Quantity"].sum()

best_city = city_sales.idxmax()
# Display the result
print(f"City sold the most products: {best_city}")
```

Output:



4.2.4. Most Frequently Sold Product Pairs:

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Code:

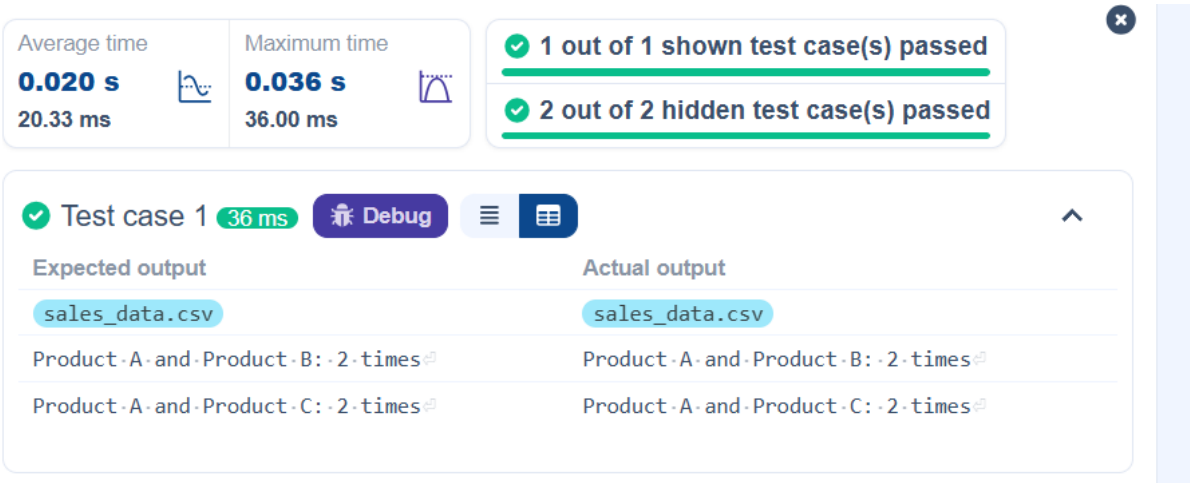
```
import pandas as pd
from itertools import combinations
from collections import Counter

# Prompt user to input the file name
file_name = input()

# Read data from the specified CSV file
df = pd.read_csv(file_name)

# write the code
date_products = {}

for date, group in df.groupby('Date'):
```

```
    products = group['Product'].unique()
    if len(products) > 1:
        date_products[date] = products
grouped = df.groupby("Date")["Product"].apply(list)
#GENERATE ALL PRODUCT PAIRS FOR EACH
CONSTRUCTIONS
pair_counter = Counter()

for products in date_products.values():
    pairs = combinations(sorted(products), 2)
#sort to ensure consistency
    pair_counter.update(pairs)

    #find the most common products pairs
if pair_counter:
    max_count = max(pair_counter.values())

# Output the most frequent product pairs
    for pair, count in pair_counter.items():
        if count == max_count:
            print(f"{pair[0]} and {pair[1]}: {count} times")
```

else:

```
    print("No product pairs found.")
```

Output:



4.2.5. Titanic Dataset Analysis and Data Cleaning:

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using .info()).
5. Get basic statistics (mean, standard deviation,

etc.) of the dataset using .describe().

6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the ‘Age’ column with the median age.
8. Fill missing values in the ‘Embarked’ column with the most frequent value (mode()).
9. Drop the ‘Cabin’ column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	F

Code:

```
import pandas as pd

import numpy as np


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')
```

```
# 1. Display the first 5 rows of the dataset
print(data.head())

# 2. Display the last 5 rows of the dataset
print(data.tail())

# 3. Get the shape of the dataset
print(data.shape)

# 4. Get a summary of the dataset (info)
print(data.info())

# 5. Get basic statistics of the dataset
print(data.describe())

# 6. Check for missing values
print(data.isnull().sum())

# 7. Fill missing values in the 'Age' column with the
median age
```

```
data['Age'].fillna(data['Age'].median(), inplace=True)

# 8. Fill missing values in the ‘Embarked’ column with the mode

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

# 9. Drop the ‘Cabin’ column due to many missing values
data.drop(columns=['Cabin'], inplace=True)

# 10. Create a new column 'FamilySize’ by adding ‘SibSp' and ‘Parch'
data['FamilySize'] = data['SibSp'] + data['Parch']
```

Output:

Test case 1 172 ms Debug

Expected output

```
   PassengerId  Survived  Pclass       F
are Cabin  Embarked
0           1         0       3       7.2
500   NaN         S
1           2         1       1      71.2
833   C85         C
2           3         1       3       7.9
250   NaN         S
3           4         1       1      53.1
000  C123         S
4           5         0       3       8.0
500   NaN         S

[5 rows x 12 columns]
     PassengerId  Survived  Pclass       F
are Cabin  Embarked
886         887         0       2      1
3.00   NaN         S
887         888         1       1      3
```

Actual output

```
   PassengerId  Survived  Pclass
 Fare Cabin  Embarked
0           1         0       3
7.2500   NaN         S
1           2         1       1      7
1.2833   C85         C
2           3         1       3
7.9250   NaN         S
3           4         1       1      5
3.1000  C123         S
4           5         0       3
8.0500   NaN         S

[5 rows x 12 columns]
     PassengerId  Survived  Pclass
 Fare Cabin  Embarked
886         887         0       2
13.00   NaN         S
887         888         1       1
```

4.2.6. Titanic Dataset Analysis and Data Cleaning – 2:

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column ‘IsAlone’ which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the ‘Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the ‘Embarked’ column, dropping the first category.
4. Get the mean age of passengers.

5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	F

Code:

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']

# 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)
```

```
data['IsAlone'] = np.where(data['FamilySize'] == 0, 1, 0)

# 2. Convert ‘Sex' to numeric (male: 0, female: 1)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

# 3. One-hot encode the ‘Embarked' column
data = pd.get_dummies(data, columns=['Embarked'])

# 4. Get the mean age of passengers
mean_age = data['Age'].mean()
print(mean_age)

# 5. Get the median fare of passengers
median_fare = data['Fare'].median()
print(median_fare)

# 6. Get the number of passengers by class
print( data['Pclass'].value_counts()
)
# 7. Get the number of passengers by gender
print( data['Sex'].value_counts())
```

```
# 8. Get the number of passengers by survival status
print( data['Survived'].value_counts())
# 9. Calculate the survival rate
survival_rate = data['Survived'].mean()
print(format(survival_rate))


# 10. Calculate the survival rate by gender
survival_by_gender = data.groupby('Sex')
['Survived'].mean()
print( survival_by_gender)
```

Output:

Average time
0.089 s
89.00 ms

Maximum time
0.089 s
89.00 ms

1 out of 1 shown test case(s) passed

Test case 1 89 ms Debug

Expected output	Actual output
29.69911764705882	29.69911764705882
14.4542	14.4542
3 491	3 491
1 216	1 216
2 184	2 184
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
0 577	0 577
1 314	1 314
Name: Sex, dtype: int64	Name: Sex, dtype: int64
0 549	0 549
1 342	1 342
Name: Survived, dtype: int64	Name: Survived, dtype: int64
0.3838383838383838	0.3838383838383838
Sex	Sex
0 0.188908	0 0.188908
1 0.742038	1 0.742038

Debugger

4.2.7. Titanic Dataset Analysis and Data Cleaning – 3:

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size

(FamilySize).

4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	F

Code:

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']
```

```
data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True)

# 1. Calculate the survival rate by class
print(data.groupby('Pclass') ['Survived'].mean())

# 2. Calculate the survival rate by embarked location
print(data.groupby ('Embarked_S') ['Survived'].mean())

# 3. Calculate the survival rate by family size
print(data.groupby('FamilySize') ['Survived'].mean())

# 4. Calculate the survival rate by being alone
print(data.groupby('IsAlone') ['Survived'].mean())

# 5. Get the average fare by class
print(data.groupby('Pclass') ['Fare'].mean())

# 6. Get the average age by class
print(data.groupby('Pclass') ['Age'].mean())
```

```
# 7. Get the average age by survival status
print(data.groupby('Survived')['Age'].mean())

# 8. Get the average fare by survival status
print(data.groupby('Survived') ['Fare'].mean())

# 9. Get the number of survivors by class
print(data[data['Survived'] == 1] ['Pclass'].value_counts())

# 10. Get the number of non-survivors by class
print(data[data['Survived'] == 0] ['Pclass'].value_counts())
```

Output:

Average time
0.096 s
96.00 ms

Maximum time
0.096 s
96.00 ms

1 out of 1 shown test case(s) passed

Test case 1 96 ms Debug

Expected output	Actual output
Pclass	Pclass
1 0.629630	1 0.629630
2 0.472826	2 0.472826
3 0.242363	3 0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S
0 0.506073	0 0.506073
1 0.336957	1 0.336957
Name: Survived, dtype: float64	Name: Survived, dtype: float64
FamilySize	FamilySize
0 0.303538	0 0.303538
1 0.552795	1 0.552795

Terminal Test cases

4.2.8. Titanic Dataset Analysis and Data Cleaning – 4:

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).

5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	F

Code:

```
import pandas as pd

import numpy as np


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True)
```

```
# 1. Get the number of survivors by gender
survivors_by_gender = data[data['Survived'] == 1]
['Sex'].value_counts()
print(survivors_by_gender)
# 2. Get the number of non-survivors by gender
non_survivors_by_gender = data[data['Survived'] == 0]
['Sex'].value_counts()
print(non_survivors_by_gender)
# 3. Get the number of survivors by embarked location
survivors_by_embarked_s = data[data['Survived'] == 1]
['Embarked_S'].value_counts()
print(survivors_by_embarked_s)

# 4. Get the number of non-survivors by embarked
location
non_survivors_by_embarked_s = data[data['Survived']
== 0]['Embarked_S'].value_counts()
print(non_survivors_by_embarked_s)
# 5. Calculate the percentage of children (Age < 18)
who survived
children = data[data['Age'] < 18]
children_survival_rate = children['Survived'].mean()
print(children_survival_rate)
```

```
# 6. Calculate the percentage of adults (Age >= 18) who survived
adults = data[data['Age'] >= 18]
adults_survival_rate = adults['Survived'].mean()
print(adults_survival_rate)

# 7. Get the median age of survivors
median_age_survivors = data[data['Survived'] == 1]['Age'].median()
print(median_age_survivors)
# 8. Get the median age of non-survivors
median_age_non_survivors = data[data['Survived'] == 0]['Age'].median()
print(median_age_non_survivors)
# 9. Get the median fare of survivors
median_fare_survivors = data[data['Survived'] == 1]['Fare'].median()
print(median_fare_survivors)
# 10. Get the median fare of non-survivors
median_fare_non_survivors = data[data['Survived'] == 0]['Fare'].median()
```

print(median_fare_non_survivors)

Output:

Average time
0.116 s
116.00 ms

Maximum time
0.116 s
116.00 ms

1 out of 1 shown test case(s) passed

Test case 1 116 ms Debug

Expected output	Actual output
female 233	female 233
male 109	male 109
Name: Sex, dtype: int64	Name: Sex, dtype: int64
male 468	male 468
female 81	female 81
Name: Sex, dtype: int64	Name: Sex, dtype: int64
1 217	1 217
0 125	0 125
Name: Embarked_S, dtype: int64	Name: Embarked_S, dtype: int64
1 427	1 427
0 122	0 122
Name: Embarked_S, dtype: int64	Name: Embarked_S, dtype: int64
0.5398230088495575	0.5398230088495575
0.3810316139767055	0.3810316139767055
28.0	28.0
28.0	28.0